

MATHS

classmate

Date

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Section - D

32. $m = \tan A + \sin A$, $n = \tan A - \sin A$

$$m^2 - n^2 = (m+n)(m-n)$$

$$= (\tan A + \sin A + \tan A - \sin A)(\tan A + \sin A - \tan A + \sin A)$$

$$= (2 \tan A)(2 \sin A)$$

$$= 4 \tan A \sin A = 4 \sqrt{\tan^2 A \sin^2 A}$$

$$= 4 \sqrt{\tan^2 A (1 - \cos^2 A)} = 4 \sqrt{\tan^2 A - \tan^2 A \cos^2 A}$$

$$= 4 \sqrt{\tan^2 A - \sin^2 A} = 4 \sqrt{(\tan A + \sin A)(\tan A - \sin A)}$$

$$= 4 \sqrt{mn}$$

33.

$$x - y + 1 = 0$$

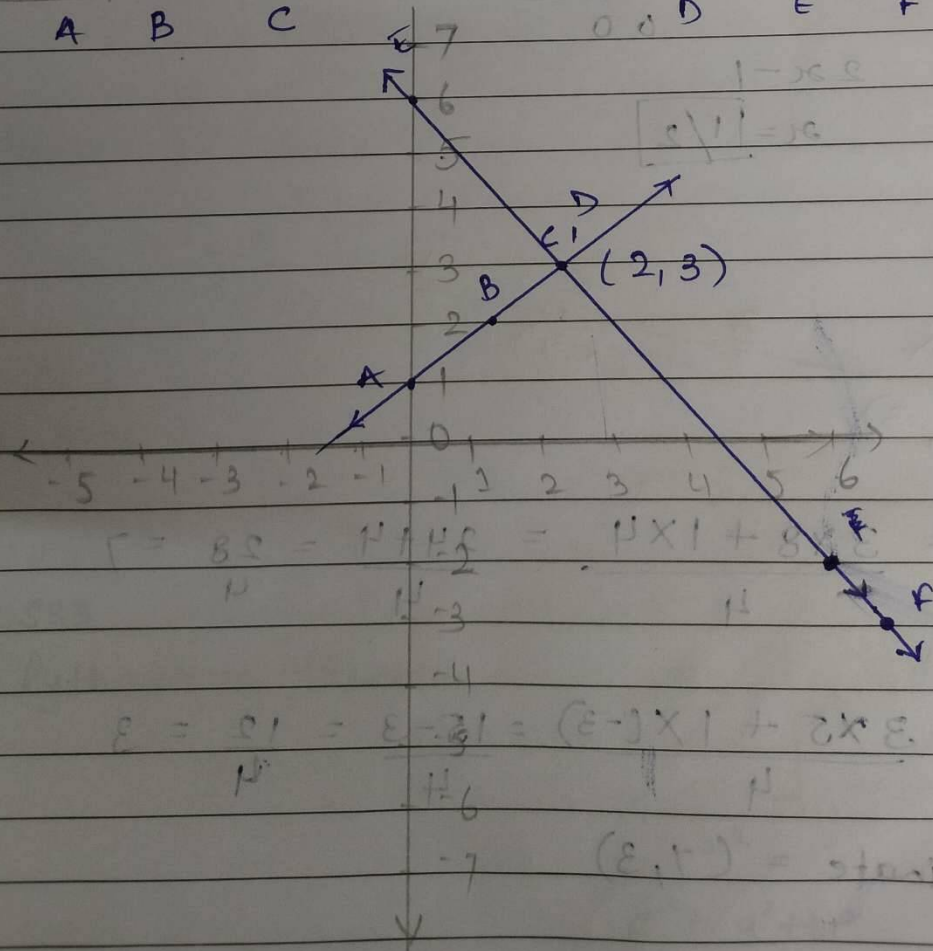
$$x = y - 1$$

x	0	1	2
y	1	2	3
	A	B	C

$$3x + 2y - 12 = 0$$

$$x = 12 - 2y / 3$$

x	2	0	6
y	3	6	-3
	D	E	F



$$\text{Area} = \frac{1 \times 3^3}{2} = 9 \text{ sq. units}$$

34. Let the third zero be α

$$(x - \sqrt{2})(x + \sqrt{2})(x + \alpha) = 2x^3 - 4x - x^2 + 2$$

$$\text{L.H.S} = (x^2 - 2)(x + \alpha)$$

$$x^3 + \alpha x^2 - 2x - 2\alpha = 2x^3 - 4x - x^2 + 2$$

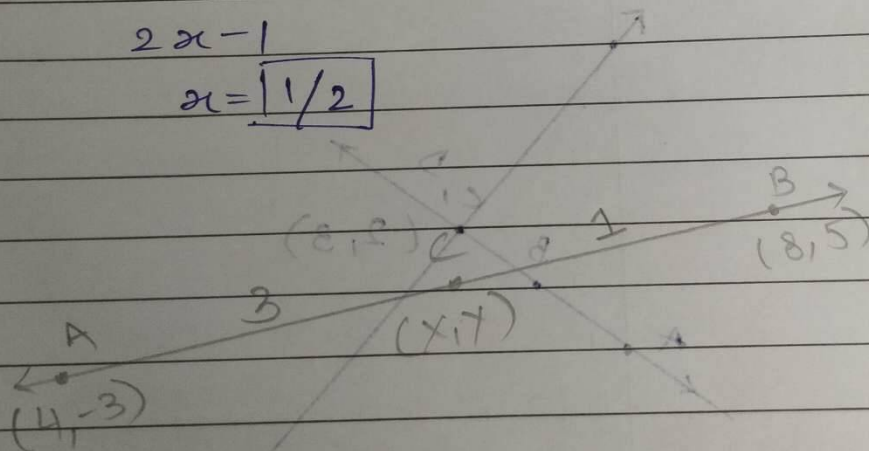
$$(x + \alpha) = (2x^3 - 4x - x^2 + 2) / (x^2 - 2)$$

$$\begin{array}{r} 2x+1 \\ x^2-2 \overline{) 2x^3-x^2-4x+2} \\ \underline{-2x^3+4x} \\ -x^2+2 \\ \underline{+x^2-2} \\ 0 \end{array}$$

$$2x-1$$

$$x = \boxed{1/2}$$

35.



$$X = \frac{3 \times 8 + 1 \times 4}{4} = \frac{24 + 4}{4} = \frac{28}{4} = 7$$

$$Y = \frac{3 \times 5 + 1 \times (-3)}{4} = \frac{15 - 3}{4} = \frac{12}{4} = 3$$

$$\text{Coordinate} = (7, 3)$$

Sec E

36. i) (2) (0, 12)

ii) (1) (3, 5)

iii) P(3, 5) Q(6, 9)

$$D = \sqrt{(3-6)^2 + (5-9)^2}$$

$$= \sqrt{(-3)^2 + (-4)^2}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$= 5m$$

(3) 5m

37. (i) (3) Parabola

(ii) Polynomial

$$(x+3)(x-2)$$

$$x^2 - 2x + 3x - 6$$

$$x^2 + x - 6$$

If $x = -3$, then $x^2 + x - 6 = (-3)^2 + (-3) - 6$

$$= 9 - 3 - 6$$

$$9 - 9 = 0$$

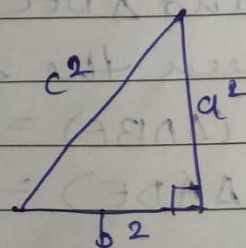
(3) 0

ii) (3) -3, 2

38. (i) (3) 24 cm²

(ii) (4) SSS

iii) (1) Pythagoras theorem



$$c^2 = a^2 + b^2$$

27. Basic Proportionality Theorem

If a line is drawn parallel to one side of a triangle intersecting the other two sides, then it divides the two sides in the same ratio.

Given: In $\triangle ABC$, $DE \parallel BC$
 DE intersects AB at D & AC at E

To prove: $\frac{AD}{DB} = \frac{AE}{EC}$

Construction: Join BE , CD & draw $EF \perp BA$ & $DG \perp CA$

Proof: Area ($\triangle ADE$) = $\frac{1}{2} \times AD \times EF$

$$\text{Area } (\triangle DBE) = \frac{1}{2} \times DB \times EF \quad \text{--- (i)}$$

$$\text{Area } (\triangle ADE) = \frac{1}{2} \times AE \times DG$$

$$\text{Area } (\triangle DEC) = \frac{1}{2} \times EC \times DG \quad \text{--- (ii)}$$

$\triangle DBE$ and $\triangle DEC$ are on the same base and between the same parallel so,

$$\text{Area } (\triangle DBE) = \text{Area } (\triangle DEC)$$

$$\text{Area } (\triangle ADE) = \text{Area } (\triangle DBE)$$

$$\text{Therefore } \frac{AD}{DB} = \frac{AE}{EC} \quad (\text{from (i) \& (ii)})$$

Q8 Let the fixed charge be x
And additional charge be y
ATQ,

$$x + 4y = 22 \quad \text{--- (i)}$$

$$x + 2y = 16 \quad \text{--- (ii)}$$

Subtract (i) from (ii)

$$\begin{array}{r} x + 2y = 16 \\ - (x + 4y = 22) \\ \hline -2y = -4 \end{array}$$

$$y = 2$$

$$x = 12$$

fixed charge = £12

Additional charge = £2

30

$$\begin{array}{r}
 15 \\
 861 \overline{) 1353} \\
 \underline{- 861} \\
 4923 \\
 \underline{- 4305} \quad 1 \\
 618
 \end{array}$$

$$\begin{array}{r}
 618 \overline{) 861} \\
 \underline{618} \\
 243
 \end{array}$$

$$\begin{array}{r}
 243 \overline{) 618} \\
 \underline{486} \\
 132
 \end{array}$$

$$\begin{array}{r}
 132 \overline{) 243} \\
 \underline{132} \\
 111
 \end{array}$$

$$\begin{array}{r}
 111 \overline{) 132} \\
 \underline{111} \\
 21
 \end{array}$$

$$\begin{array}{r}
 21 \overline{) 111} \\
 \underline{105} \\
 6
 \end{array}$$

$$\begin{array}{r}
 6 \overline{) 21} \\
 \underline{18} \\
 3
 \end{array}$$

$$\begin{array}{r}
 3 \overline{) 6} \\
 \underline{6} \\
 0
 \end{array}$$

$$HCF = 3$$

Section B

$$\begin{aligned}
 \text{Q1. } & 6 \times 5 \times 4 \times 3 \times 2 \times 1 + 5 \\
 & 5(6 \times 4 \times 3 \times 2 \times 1 + 1) \\
 & 5(145)1
 \end{aligned}$$

It has more than two factors
that's why.

$$\text{Q2. } \frac{4}{k} = \frac{6}{18} \cdot \frac{1}{3} \neq \frac{7}{21}$$

$$12 = k$$

Q3

23. Origin $(0,0)$ $(-6,8)$

$$D = \sqrt{(0+6)^2 + (0-8)^2}$$

$$= \sqrt{36+64}$$

$$= \sqrt{100}$$

$$= 10 \text{ units}$$

24. $\sin 3A = \cos (A-10^\circ)$

$$\cos (90^\circ - 3A) = \cos (A-10^\circ)$$

$$90^\circ - 3A = A - 10^\circ$$

$$100^\circ = 4A$$

$$A = 25^\circ$$

$$A = 25^\circ$$

25. In $\triangle AOB$ & $\triangle DOC$

$\angle AOB = \angle COD$ (Vertically opposite angles)

~~AB~~

$AB \parallel CD$ (given) & AD is transversal then,

$\angle BAD = \angle BDC$ (Alternate interior angles)

Thus, by AA similarity it's proved that

$$\triangle AOB \sim \triangle DOC$$

Section - A

- 1) (3) 3
- 2) (3) $\tan 60^\circ$
- 3) (4) 100°
- 4) (3) $20/3 \text{ cm}$
- 5) (4) $a=0$
- 6) (4) four decimal places
- 7) (3) 3 and 1
- 9) (3) infinitely many solutions

10) (1) $k = -9/2$

11) (1) $k \neq 3$

12) (1) 8

13) (1) Similar

14) (1) $x^3 y^4 z^4$

15) (3) $x^2 - x - 6$

16) (3) 2 intersecting or coincident

17) (1) parallel

18) (4) 16:81

19) (3) A is true but R is false

20) (3) A is true but R is false

8) ~~(1)~~ (4) (1,0)

31. Let the coordinate be (x, y)

$$\begin{aligned} & \sqrt{(5-x)^2 + (-2-y)^2} \\ & \sqrt{25 + x^2 - 10x + 4 + y^2 + 4y} \\ & \sqrt{29 + x^2 + y^2 + 4y - 10x} \quad \text{--- (i)} \end{aligned}$$

$$\begin{aligned} & \sqrt{(-3-x)^2 + (2-y)^2} \\ & \sqrt{9 + x^2 + 6x + 4 + y^2 - 4y} \\ & \sqrt{13 + x^2 + y^2 + 6x - 4y} \quad \text{--- (ii)} \end{aligned}$$

Equating (i) & (ii) & squaring (i) & (ii)

$$\begin{aligned} 29 + x^2 + y^2 + 4y - 10x &= 13 + x^2 + y^2 + 6x + \cancel{4y} - 4y \\ 16 &= 16x \\ x &= 1 \end{aligned}$$

$$\begin{aligned} 16 &= 16 + 8y = 16x \\ 16x - 8y - 16 &= 0 \\ 2x - y - 2 &= 0 \\ 2x - 2 &= y \end{aligned}$$

$$x = 1$$

$$y = 0$$

$$(1, 0)$$